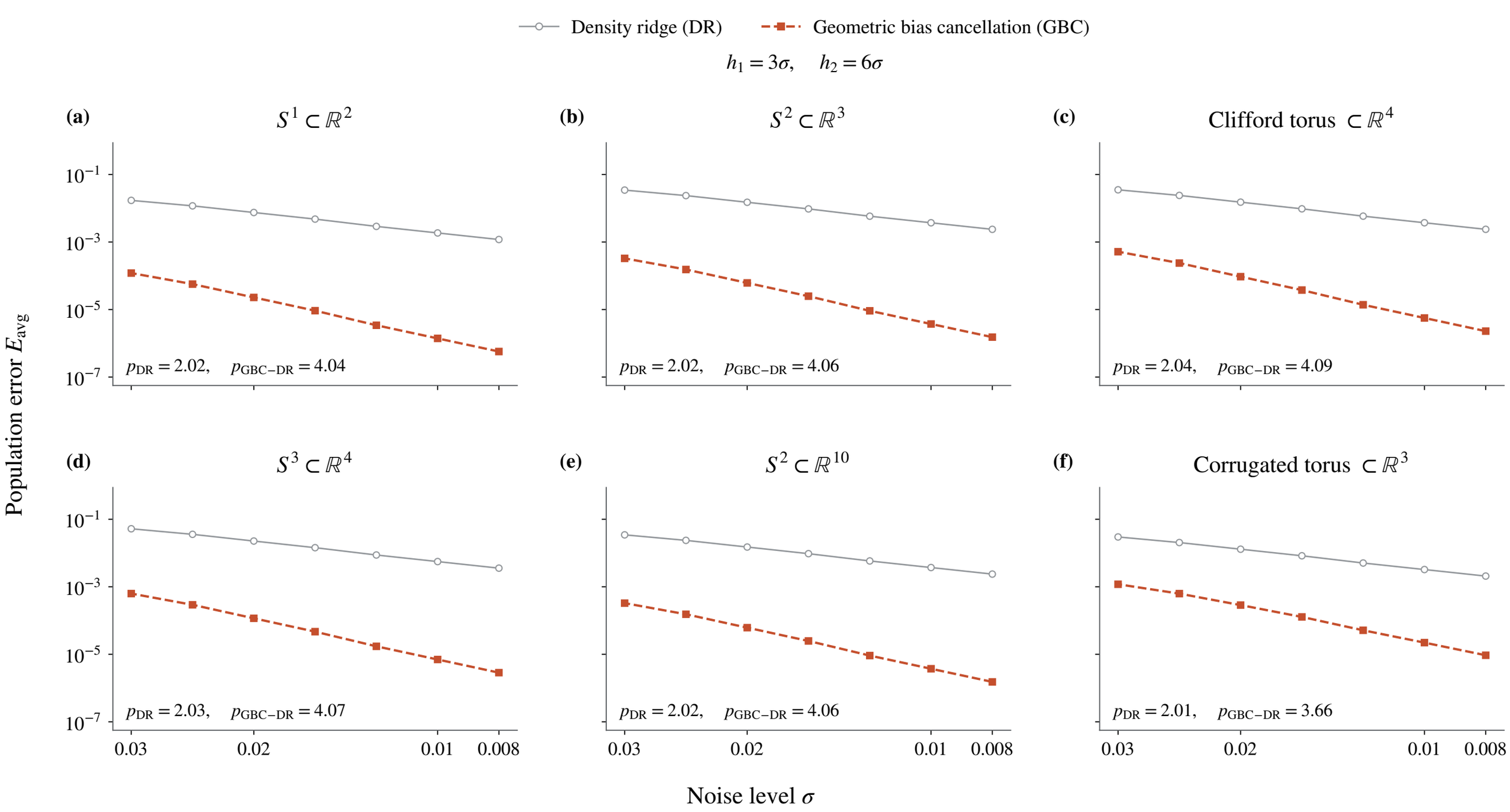
Density ridge (DR)
Geometric bias cancellation (GBC)
$h_1 = 3\sigma, \quad h_2 = 6\sigma$
(a)
$S^1 \subset \mathbb{R}^2$
(b)
$S^2 \subset \mathbb{R}^3$
(c)
Clifford torus $\subset \mathbb{R}^4$
(d)
$S^3 \subset \mathbb{R}^4$
(e)
$S^2 \subset \mathbb{R}^{10}$
(f)
Corrugated torus $\subset \mathbb{R}^3$
Population error E_{avg}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
$p_{\text{DR}} = 2.02, \quad p_{\text{GBC-DR}} = 4.04$
$p_{\text{DR}} = 2.02, \quad p_{\text{GBC-DR}} = 4.06$
$p_{\text{DR}} = 2.04, \quad p_{\text{GBC-DR}} = 4.09$
$p_{\text{DR}} = 2.03, \quad p_{\text{GBC-DR}} = 4.07$
$p_{\text{DR}} = 2.02, \quad p_{\text{GBC-DR}} = 4.06$
$p_{\text{DR}} = 2.01, \quad p_{\text{GBC-DR}} = 3.66$
0.03
0.02
0.01
0.008
Noise level σ